CS2316 Final Project
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COAL AND EMISSIONS

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POINT OF INTEREST AND PURPOSE

The purpose of this project is to analyze the impact of coal mining emissions on global greenhouse gas levels, with a focus on understanding the extent and trends of emissions generated by coal-dependent sectors. Coal mining plays a significant role in contributing to climate change through the release of CO_2. By examining data related to emissions and sectoral contributions, this analysis aims to provide a clearer picture of coal mining's role in exacerbating climate change. Furthermore, the project will explore the potential for emission reduction strategies within coal-dependent sectors to mitigate their environmental impact.

I chose this topic because climate change is one of the most pressing global issues, with profound consequences. Coal mining remains a critical energy source, heavily used in many countries despite its environmental impact. Understanding the specific contribution of coal mining to greenhouse gas levels is essential for developing targeted policies and transition plans. Through this project, I hope to uncover actionable insights that can inform efforts to reduce emissions and support a global shift toward cleaner, more sustainable energy solutions.

My hypothesis was that countries with more coal mining operations release a larger amount of CO_2 which causes an increase in temperatures (global warming).

HYPOTHESIS

PHASE I: DATA COLLECTION

CSV: I found my CSV file on a website called “climatetrace.org.” This dataset was 4.3 MB. This large dataset has information that includes what countries and companies are coal mining for fossil fuel operations. I chose this dataset because it focused on fossil-fuel operations, but specifically coal mining. This dataset helps to provide insights on coal mining operations.

HTML: I found my HTML site to scrape during Phase II after the previous website turned out to be unhelpful. This website includes CO2 emissions, Population (2022), per capita, share of world, and 1 Year Change information on every country. This data helps quantify each country’s contribution to global CO2 emissions.

API: I also found the API during Phase II once I realized the previous API was not helpful. I found it on the “climatewatchdata.org” website. This API includes country, year, and sector. This API provides data specific to the release of CO2 which is the greenhouse gas released from coal mining operations.

PHASE II: DATA CLEANING

CSV: One inconsistency in the dataset was the presence of empty columns that contained NaN values, which indicate missing or unavailable data. These NaN values are significant as they can disrupt analyses. I handled this issue by using .replace() and pd.NA to fill all NaN values with 'n/a', a placeholder that clearly denotes missing data in a readable format. The API and the downloaded datasets have differing date formats, this inconsistency is significant because varying date formats complicate sorting and make it challenging to merge datasets, leading to errors in analyses. To handle this, I reformatted the dates in the downloaded datasets to match the YYYY format used by the API and website.

HTML: The web page has the full names of the countries but the dowloaded dataset has the acronym versions. In order to merge them I need to make sure the country names are uniform. So I changed the full names to the acronyms to match the web collection data to the downloaded dataset.

API: An inconsistency I discovered in the API was the difference in units for CO_2 emissions data. The API provided CO_2 emissions in megatonnes ($MtCO_2e$), while the second web collection source measured emissions in tons. This difference is significant because it could lead to inaccurate comparisons or analyses if the units aren't aligned. To resolve this, I converted all CO_2 emissions values from megatonnes to tons by multiplying each value by 1,000,000, ensuring consistency across the datasets.

INSIGHT #1

- took in two csv files to find correlation between coal mining operations and CO2 emissions among the top contributing countries
- got CO2 emissions data from the first file and the number of coal mining operations from the second file
- used new function: nlargest() which returns a specified number of rows, in this case 5, for the specified column: CO2 Emissions (tons, 2022)
- this comparison emphasizes the strong link between coal dependency and emissions

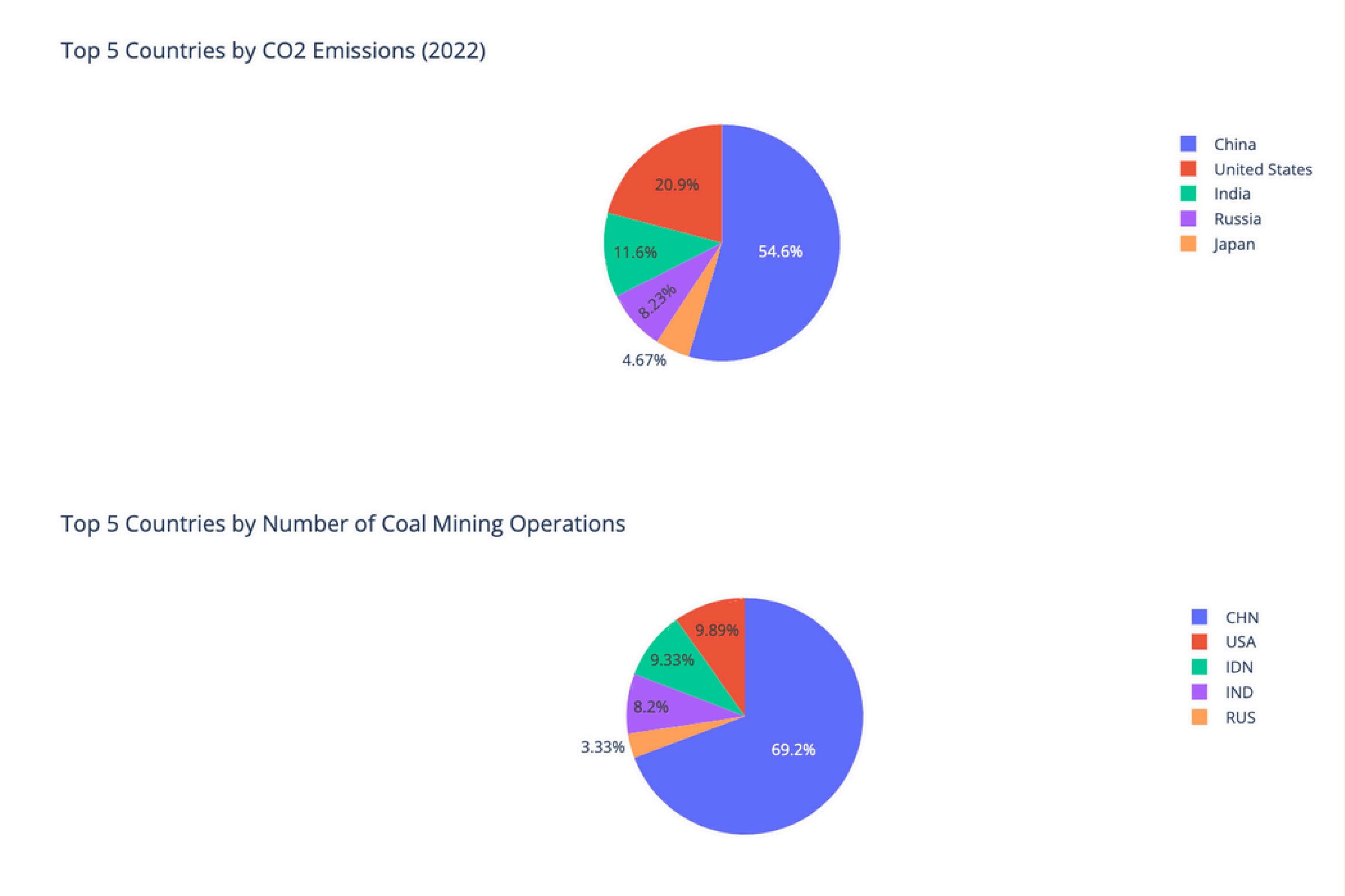
```
#find top 5 countries by CO2 emissions
top_emissions = country_emissions.nlargest(5, 'CO2 Emissions (tons, 2022)')
print("Top 5 countries by CO2 emissions:")
print(top_emissions[['Country', 'CO2 Emissions (tons, 2022)']])
```

```
#count the number of coal mining operations per country
country_counts = data['iso3_country'].value_counts().reset_index()
country_counts.columns = ['country', 'num_operations']
```

```
#find top 5 countries by coal mining operations
top_countries = country_counts.head(5)
print("Top 5 countries by number of coal mining operations:")
print(top_countries)

return top_emissions, top_countries
```

```
Top 5 countries by CO2 emissions:
         Country  CO2 Emissions (tons, 2022)
0          China                 12667428430
1  United States                  4853780240
2          India                  2693034100
3         Russia                  1909039310
4          Japan                  1082645430
Top 5 countries by number of coal mining operations:
  country  num_operations
0     CHN            3368
1     USA             481
2     IDN             454
3     IND             399
4     RUS             162
```

VISUALIZATION #1




```python
def visual1(top_emissions, top_countries):
    #visualization for CO2 emissions
    fig1 = px.pie(
        top_emissions,
        names='Country',
        values='CO2 Emissions (tons, 2022)',
        title='Top 5 Countries by CO2 Emissions (2022)',
        labels={'CO2 Emissions (tons, 2022)': 'CO2 Emissions (tons)'}
    )
    fig1.show()

    #visualization for coal mining operations
    fig2 = px.pie(
        top_countries,
        names='country',
        values='num_operations',
        title='Top 5 Countries by Number of Coal Mining Operations',
        labels={'num_operations': 'Number of Operations'}
    )
    fig2.show()
```

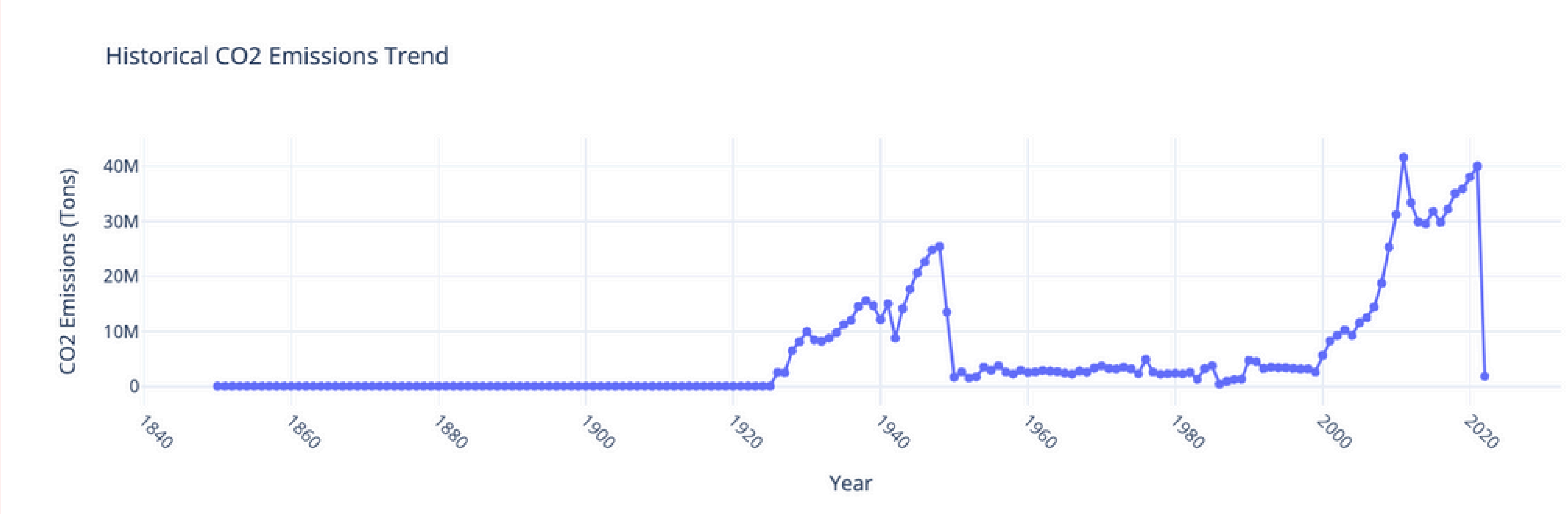
INSIGHT #2

- analyzes the efficiency of coal mining operations by calculating the tons of CO2 emissions per mining operation for each country by taking in the same two csv files
- used two new functions: map() to manually map ISO3 codes to full country names and merge() to combine the emissions and coal data and create a new dataframe

	Country	CO2 Emissions (tons, 2022)	num_operations	tons_emissions_per_operation
0	China	12667428430	3368	3.761113e+06
1	United States	4853780240	481	1.009102e+07
2	India	2693034100	399	6.749459e+06
3	Russia	1909039310	162	1.178419e+07
4	Australia	393162550	148	2.656504e+06

```
#count the number of coal mining operations per country
coal_counts = coal_operations['iso3_country'].value_counts().reset_index()
coal_counts.columns = ['country', 'num_operations']

#manual mapping of ISO3 codes to full country names
iso3_to_name = {
    'USA': 'United States',
    'CHN': 'China',
    'IND': 'India',
    'AUS': 'Australia',
    'RUS': 'Russia',
}
coal_counts['country_full'] = coal_counts['country'].map(iso3_to_name)

#merge the two datasets
merged_data = pd.merge(
    country_emissions,
    coal_counts,
    left_on='Country',
    right_on='country_full',
    how='inner'
)

#calculate emissions per operation
merged_data['tons_emissions_per_operation'] = (
    merged_data['CO2 Emissions (tons, 2022)'] / merged_data['num_operations']
)
```

INSIGHT #3




```
: total_emissions    9.788500e+08
  mean_emissions     1.166687e+06
  emission_count     8.390000e+02
  Name: CO2 Emissions In Tons, dtype: float64
```

```python
#grouping data by year and summing emissions
emissions_trend = weather_data.groupby('year')['CO2 Emissions In Tons'].sum().reset_index()

#plotting the historical CO2 emissions trend using Plotly
fig = px.line(
    emissions_trend,
    x='year',
    y='CO2 Emissions In Tons',
    title='Historical CO2 Emissions Trend',
    labels={'year': 'Year', 'CO2 Emissions In Tons': 'CO2 Emissions (Tons)'},
    markers=True
)
```

```python
    #finding emissions data
    global_emissions_summary = weather_data['CO2 Emissions In Tons'].agg(
    total_emissions='sum',
    mean_emissions='mean',
    emission_count='count'
)
```

- the dataset provides CO_2 emission data by year, which allows for analyzing long-term trends in emissions.
- used agg() to get the total emissions, mean emissions, and emissions count
- historical CO_2 emissions trend highlights significant increases in emissions over time

INSIGHT #4

- calculates the total amount of emissions released by each sector
- used new functions idxmin()/idxmax() which provides the index label of the min/max value in a Series or DataFrame
- reveals that the Energy sector is the most significant contributor to CO2 emissions
- Land-Use Change and Forestry sector emerges as the least emitting sector
- Addressing emissions in this sector presents the greatest opportunity for impactful climate action

```
#aggregate total emissions by sector
total_emissions_by_sector = data.groupby('sector')['CO2 Emissions In Tons'].sum()

#aggregate average emissions by sector
avg_emissions_by_sector = data.groupby('sector')['CO2 Emissions In Tons'].mean()

#identify the most and least emitting sectors
most_emitting_sector = total_emissions_by_sector.idxmax()
least_emitting_sector = total_emissions_by_sector.idxmin()
```

```
{'Total Emissions by Sector': {'Building': 25700000,
  'Bunker Fuels': 3600000,
  'Electricity/Heat': 6380000,
  'Energy': 416909999,
  'Industrial Processes': 1070000,
  'Industrial Processes and Product Use': 110000,
  'Land-Use Change and Forestry': -25440000,
  'Manufacturing/Construction': 52720000,
  'Other Fuel Combustion': 1350000,
  'Total excluding LUCF': 165500000,
  'Total excluding LULUCF': 252559999,
  'Transportation': 78390000},
 'Most Emitting Sector': {'Sector': 'Energy'},
 'Least Emitting Sector': {'Sector': 'Land-Use Change and Forestry'}}
```

INSIGHT #5

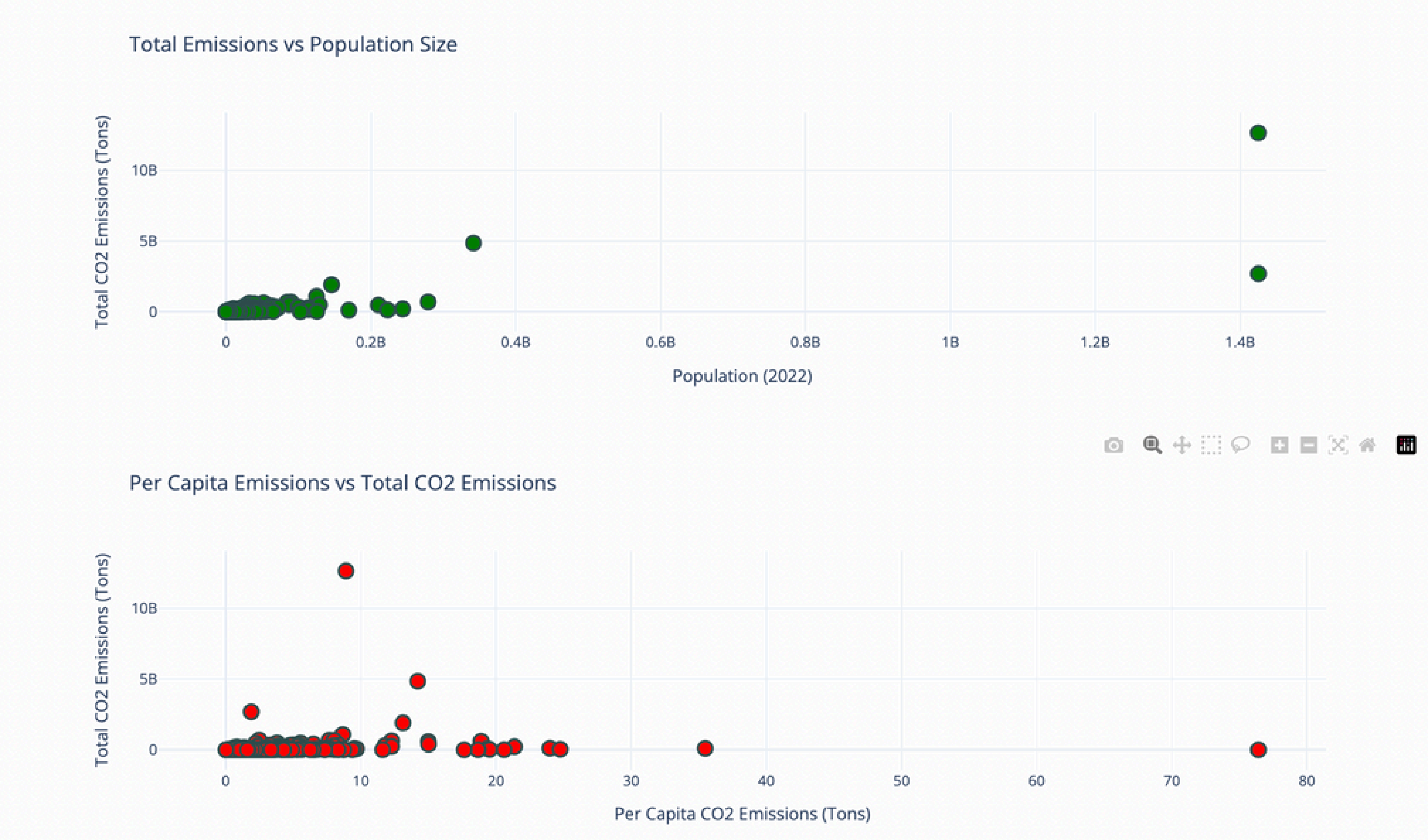
	company_name	Share (%)
0	Eastern Coalfields	1.347414
1	South Eastern Coalfields	1.074290
2	Western Coalfields	1.019665
3	The Singareni Collieries	0.855790
4	Heilongjiang Longmay MINING Group	0.819374
...	...	...
1985	International Prima Coal	0.018208
1986	Indomarta Multi Mining	0.018208
1987	Intan Karya Mandiri	0.018208
1988	Indomas Karya Jaya	0.018208
1989	Zambezi Gas Zimbabwe	0.018208

1990 rows × 2 columns

```
#calculate the total entries for each company
company_shares = data['company_name'].value_counts()

#calculate the percentage share of each company
total_entries = company_shares.sum()
company_shares_percentage = (company_shares / total_entries) * 100

#result as a DataFrame
company_shares_df = company_shares_percentage.reset_index(name='Share (%)').rename(columns={'index': 'Company Name'})

return company_shares_df
```

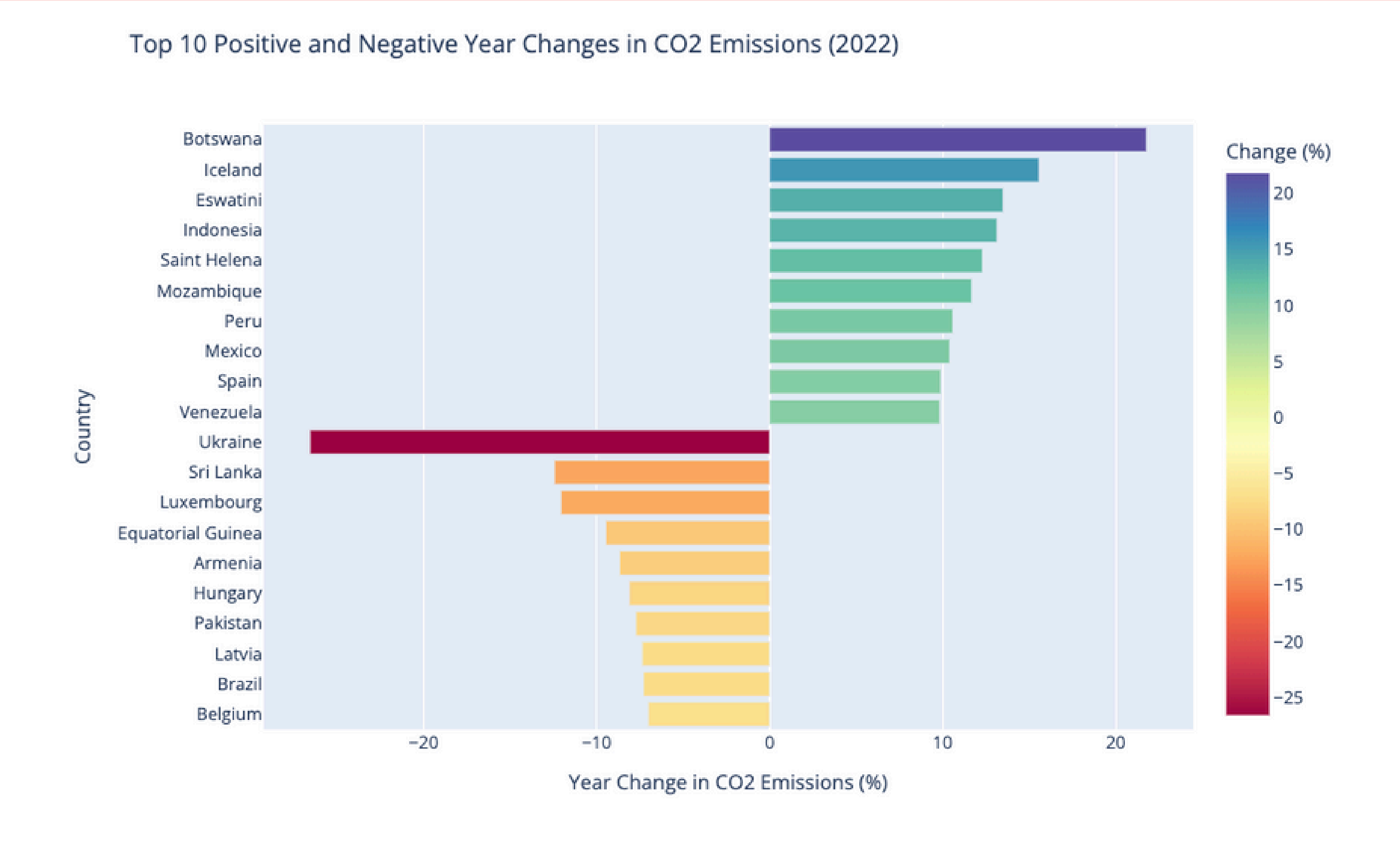
- analyzes each company's share in coal mining operations
- used value_counts() and sum() to calculate the percentange share of each company to create a new dataframe with the data
- nearly two thousand companies make up for all the coal mining operations and working with the one's with the biggest share would best help address the release of CO2 emissions

VISUALIZATION #2




```python
def visual2(emissions):

    #load the cleaned dataset
    co2_emissions = pd.read_csv(emissions)

    #analyze relationship between population size and emissions (per capita vs total emissions)
    co2_emissions = co2_emissions[['Country', 'CO2 Emissions (tons, 2022)', 'Population (2022)', 'Per capita']]
    co2_emissions = co2_emissions.dropna()

    #convert numeric columns to appropriate types
    co2_emissions['CO2 Emissions (tons, 2022)'] = co2_emissions['CO2 Emissions (tons, 2022)'].str.replace(',', '').astype(float)
    co2_emissions['Population (2022)'] = co2_emissions['Population (2022)'].str.replace(',', '').astype(float)
    co2_emissions['Per capita'] = co2_emissions['Per capita'].astype(float)

    #plot total emissions vs population size
    fig2 = px.scatter(
        co2_emissions,
        x='Population (2022)',
        y='CO2 Emissions (tons, 2022)',
        title='Total Emissions vs Population Size',
        labels={
            'Population (2022)': 'Population (2022)',
            'CO2 Emissions (tons, 2022)': 'Total CO2 Emissions (Tons)'
        },
        template='plotly_white'
    )
    fig2.update_traces(marker=dict(size=12, color='green', line=dict(width=2, color='DarkSlateGrey')))
    fig2.show()

    #plot per capita emissions vs total emissions
    fig3 = px.scatter(
        co2_emissions,
        x='Per capita',
        y='CO2 Emissions (tons, 2022)',
        title='Per Capita Emissions vs Total CO2 Emissions',
        labels={
            'Per capita': 'Per Capita CO2 Emissions (Tons)',
            'CO2 Emissions (tons, 2022)': 'Total CO2 Emissions (Tons)'
        },
        template='plotly_white'
    )
    fig3.update_traces(marker=dict(size=12, color='red', line=dict(width=2, color='DarkSlateGrey')))
    fig3.show()
```

VISUALIATION #3




```python
def visual3(file):
    #load the dataset
    data = pd.read_csv(file)

    #fixing the "Year Change" column by removing percentage signs and converting to numeric
    data['1 Year Change'] = data['1 Year Change'].str.replace('%', '').astype(float)

    #sorting the data by "1 Year Change"
    sorted_co2_data = data.sort_values(by='1 Year Change', ascending=False)

    #limiting the dataset to the top 10 positive and top 10 negative changes - better for visualization
    top_positive_changes = sorted_co2_data.nlargest(10, '1 Year Change')
    top_negative_changes = sorted_co2_data.nsmallest(10, '1 Year Change')
    limited_data = pd.concat([top_positive_changes, top_negative_changes])

    #creating bar chart
    fig = px.bar(
        limited_data,
        x='1 Year Change',
        y='Country',
        orientation='h',
        color='1 Year Change',
        title='Top 10 Positive and Negative Year Changes in CO2 Emissions (2022)',
        labels={'1 Year Change': 'Year Change in CO2 Emissions (%)', 'Country': 'Country'},
        color_continuous_scale='Spectral',
    )

    #updating layout for better readability
    fig.update_layout(
        xaxis_title='Year Change in CO2 Emissions (%)',
        yaxis_title='Country',
        yaxis=dict(autorange="reversed"),  #reverse the y-axis for better visualization
        coloraxis_colorbar=dict(title="Change (%)"),
        width=900,  #adjusing plot width and height for better visibility
        height=600
    )
    fig.show()
```

RESULTS: CONCLUSION

Almost all of the data that I cleaned, parsed, and analyzed supported my original hypothesis. The correlation between coal mining operations and CO2 emissions found to be consistently high throughout the data. There are many applications that came from my conclusions:

- decrease coal dependency in order to reduce emissions
- China, USA, India and Russia have the most CO2 emissions and coal mining operations
    - best to work with these countries to decrease the amount of CO2 emissions’
- Energy sector is the most significant contributor to CO2 emissions
    - addressing emissions in this sector presents the greatest opportunity for impactful climate action
- nearly two thousand companies make up for all the coal mining operations
    - working with the one’s with the biggest share (Eastern Coalfields, South Eastern Coalfields, and Western Coalfield) would best help address the release of CO2 emissions

WHAT I LEARNED

Learned:

- How to find and clean real datasets with inconsistencies
- Learnt how to use new functions such as idxmin(), idxmax(), map(), and merge()
- Learned how to aggregate and summarize data using functions like groupby() and agg() to uncover trends and patterns

Challenges:

- Finding large datasets with valid and varying inconsistencies
- Finding useful applications of data after cleaning it
- Managing the trade-off between data cleaning and retaining essential information for analysis

THANK YOU